Lecture 9. Hypothesis Testing (Part 1)

One-Sample Tests and Weighted K-Sample Log-Rank Tests

Contents

1 Overview **1**

1.1 One-sample problems . . . 1

1.2 Two- and K-sample problems . . . 1

1.3 How to test? . . . 2

2 One-Sample Test with Right-Censored Data **2**

2.1 One-sample log-rank test . . . 2

2.2 Harrington–Fleming weights . . . 3

3 K-Sample Tests **3**

3.1 Hypotheses . . . 3

3.2 Data and notation . . . 3

3.3 Basic idea and the test statistic . . . 3

3.4 Distribution under H_0 . . . 4

3.5 Linear dependency and the final statistic . . . 4

3.6 Special case $K = 2$ . . . 5

4 Choice of the Weight Function **5**

5 Catheter Placement in Kidney Dialysis Patients **5**

6 Three-Group Comparison (Bone Marrow Transplant) **6**

7 Selected exercises **7**

1 Overview

1.1 One-sample problems

Rationale. A KM pattern may suggest a parametric model (efficient, handles complex sampling), whose fit must then be checked:

$$H_0 : S(t) = S_0(t), \qquad H_0 : H(t) = H_0(t), \qquad H_0 : h(t) = h_0(t) \qquad \text{for all } t \le \tau,$$

with S_0, H_0, h_0 *known*, e.g. $H_0(t) = \lambda t$ for $\text{Exp}(\lambda)$.

1.2 Two- and K-sample problems

For heterogeneous populations, compare groups:

$$H_0 : S_1(t) = S_2(t) \iff H_1(t) = H_2(t) \iff h_1(t) = h_2(t), \qquad t \in [0, \tau],$$

hypotheses of *no* known form; K samples: $H_0 : S_1(t) = \cdots = S_K(t)$.

1.3 How to test?

Without censoring: t or Kolmogorov–Smirnov tests, ANOVA, Wilcoxon or Kruskal–Wallis. The difficulty is censoring.

2 One-Sample Test with Right-Censored Data

Data $\{(T_i, \delta_i)\}$, X_i with hazard h. Test $H_0 : h(t) = h_0(t)$ (equivalently $H(t) = H_0(t)$) on $[0, \tau]$, τ typically the largest study time; t_j, d_j, Y_j as before.

Definition 9.1: One-sample weighted statistic

With a weight $W(t)$, $W(t) = 0$ when $Y(t) = 0$, compare Nelson–Aalen increments with the hypothesized hazard:

$$Z(\tau) = O(\tau) - E(\tau) = \sum_{j=1}^{D} W(t_j) \frac{d_j}{Y(t_j)} - \int_0^\tau W(s) h_0(s)\, \mathrm{d}s.$$

Under H_0

$$V[Z(\tau)] = \int_0^\tau W^2(s) \frac{h_0(s)}{Y(s)}\, \mathrm{d}s,$$

and $Z(\tau)^2 / V[Z(\tau)] \sim \chi^2_1$ approximately.

Remark 9.2: Intuition

Z is a weighted distance $\int W\, \mathrm{d}(\tilde{H} - H_0)$; each increment d_j/Y_j has (Poisson) variance $\approx h_0\, \mathrm{d}s/Y$.

- χ^2: two-sided alternatives only.
- One-sided: reject for large positive $T_1 = Z(\tau)/\sqrt{V[Z(\tau)]}$ against $h > h_0$, large negative against $h < h_0$.

2.1 One-sample log-rank test

Theorem 9.3: $W(t) = Y(t)$: observed vs. expected

With left truncation (T_i time on study, L_i entry time),

$$O(\tau) = \sum_{j=1}^{D} Y(t_j) \frac{d_j}{Y(t_j)} = \sum_{j=1}^{D} \sum_{i=1}^{n} I(T_i = t_j, \delta_i = 1) = \sum_{i=1}^{n} I(\delta_i = 1),$$

$$E(\tau) = \int_0^\tau Y(s) h_0(s)\, \mathrm{d}s = \sum_{i=1}^{n} \int_0^\tau I(L_i \le s \le T_i) h_0(s)\, \mathrm{d}s = \sum_{i=1}^{n} [H_0(T_i) - H_0(L_i)].$$

O: observed events by τ; E: expected under H_0; $V[Z(\tau)] = E(\tau)$, so the statistic is $(O - E)^2/E$.

2.2 Harrington–Fleming weights

$$W_{HF}(t) = Y(t)S_0(t)^p[1 - S_0(t)]^q, \qquad p, q \geq 0, \quad S_0(t) = \exp[-H_0(t)].$$

- $p \gg q$: **early** departures emphasized;
- $q \gg p$: **late** departures;
- $p = q$: **mid-range**;
- $p = q = 0$: the one-sample log-rank test.

Example 9.4: IBCSG Trial 6

Multi-center trial of adjuvant chemotherapy, 735 subjects. The NA cumulative hazard is roughly linear up to 100 months (exponential), slope $\hat{\lambda} \approx 0.005$. One-sample log-rank tests ($W = Y$):

H_0	$T_1 = Z(\tau)/\sqrt{V[Z(\tau)]}$	p-value
$H_0(t) = 0.0045t$	-0.002	0.998
$H_0(t) = 0.004t$	1.8	0.07
$H_0(t) = 0.005t$	-1.4	0.16
$H_0(t) = 0.006t$	-4.1	< 0.01

$0.0045t$ fits best (a larger hypothesized rate inflates E, making $O - E$ negative).

3 K-Sample Tests

3.1 Hypotheses

$H_0 : h_1(t) = h_2(t) = \cdots = h_K(t) \ \ \forall t \leq \tau$ vs. $H_A :$ at least one $h_k(t)$ differs for some $t \leq \tau$.

τ is the largest time with all groups at risk, typically $\min_k t_{\max,k}$; the alternative is *global*.

3.2 Data and notation

Data $\{(T_{ik}, \delta_{ik})\}$; $0 < t_1 < \cdots < t_D$ the distinct pooled death times:

- d_j, d_{jk}: deaths at t_j in the pooled sample / in group k;
- Y_j, Y_{jk}: at risk at t_j, pooled / in group k.

3.3 Basic idea and the test statistic

Under H_0 the pooled and within-group NA estimators estimate the same H:

$$\tilde{H}(t) = \sum_{t_j \leq t} \frac{d_j}{Y_j} \to \int_0^t h(u)\,\mathrm{d}u, \qquad \tilde{H}_k(t) = \sum_{t_j \leq t} I(Y_{jk} > 0)\frac{d_{jk}}{Y_{jk}} \to \int_0^t h(u)\,\mathrm{d}u.$$

Test a weighted difference of $\tilde{H}_k$ and $\tilde{H}$.

Definition 9.5: Weighted K-sample statistic

With a random weight $W_k(t) \geq 0$, zero when $Y_j = 0$ or $Y_{jk} = 0$:

$$Z_k(\tau) = \sum_{j=1}^{D} W_k(t_j)\Big\{\frac{d_{jk}}{Y_{jk}} - \frac{d_j}{Y_j}\Big\}.$$

Typically $W_k(t_j) = Y_{jk}W(t_j)$ with common W:

$$Z_k(\tau) = \sum_{j=1}^{D} W(t_j)\Big\{d_{jk} - Y_{jk}\frac{d_j}{Y_j}\Big\} \quad = \sum_j W(t_j)\,(\text{observed} - \text{expected deaths in group } k).$$

3.4 Distribution under H_0

Theorem 9.6: Asymptotic normality (via martingale theory)

Under H_0, $(Z_1(\tau), \ldots, Z_K(\tau)) \approx \text{MVN}(\mathbf{0}, \Sigma)$, $\Sigma = (\sigma_{kk'})_{K\times K}$, with

$$\hat\sigma_{kk} = \sum_{j=1}^{D} W(t_j)^2 \frac{Y_{jk}}{Y_j}\Big(1 - \frac{Y_{jk}}{Y_j}\Big)\frac{Y_j - d_j}{Y_j - 1}\, d_j, \qquad k = 1, \ldots, K,$$

$$\hat\sigma_{kk'} = -\sum_{j=1}^{D} W(t_j)^2 \frac{Y_{jk}Y_{jk'}}{Y_j^2}\frac{Y_j - d_j}{Y_j - 1}\, d_j, \qquad k \neq k'.$$

Derivation: hypergeometric origin of the variance

Given the margins at t_j, under H_0 the d_j deaths are a random draw from the Y_j at risk, so $(d_{j1}, \ldots, d_{jK})$ is multivariate hypergeometric (Exercise 9.1):

$$\text{E}(d_{jk}) = d_j\frac{Y_{jk}}{Y_j}, \quad \text{Var}(d_{jk}) = d_j\frac{Y_{jk}}{Y_j}\Big(1 - \frac{Y_{jk}}{Y_j}\Big)\frac{Y_j - d_j}{Y_j - 1}, \quad \text{Cov}(d_{jk}, d_{jk'}) = -d_j\frac{Y_{jk}Y_{jk'}}{Y_j^2}\frac{Y_j - d_j}{Y_j - 1}.$$

$(Y_j - d_j)/(Y_j - 1)$ is the finite-population correction (1 if $d_j = 1$). Sum over (approximately uncorrelated) death times.

3.5 Linear dependency and the final statistic

Theorem 9.7: $\sum_k Z_k(\tau) = 0$

$$\sum_{k=1}^{K} Z_k(\tau) = \sum_{j=1}^{D} W(t_j)\sum_{k=1}^{K}\Big\{d_{jk} - d_j\frac{Y_{jk}}{Y_j}\Big\} = \sum_{j=1}^{D} W(t_j)(d_j - d_j) = 0,$$

using $\sum_k d_{jk} = d_j$, $\sum_k Y_{jk} = Y_j$: Σ is singular.

Drop one $Z_K(\tau)$:

$$(Z_1, \ldots, Z_{K-1}) \approx \text{MVN}(\mathbf{0}, \Sigma_{(K-1)\times(K-1)}),$$

and

$$\chi^2 = (Z_1(\tau), \ldots, Z_{K-1}(\tau))\,\hat\Sigma^{-1}_{(K-1)\times(K-1)}\,(Z_1(\tau), \ldots, Z_{K-1}(\tau))^T \overset{H_0}{\sim} \chi^2_{K-1}.$$

Reject if $\chi^2 > \chi^2(K-1, 1-\alpha)$; the value does not depend on the dropped Z_k (Exercise 9.3).

3.6 Special case $K = 2$

$$Z = \frac{Z_1(\tau)}{\sqrt{\hat{V}(Z_1(\tau))}} = \frac{\sum_{j=1}^{D} W(t_j)(d_{j1} - Y_{j1}d_j/Y_j)}{\sqrt{\sum_{j=1}^{D} W^2(t_j)\frac{Y_{j1}}{Y_j}(1 - \frac{Y_{j1}}{Y_j})\frac{Y_j - d_j}{Y_j - 1}d_j}} \overset{H_0}{\approx} N(0,1).$$

Reject if $|Z| > Z_{1-\alpha/2}$, or use one-sided tests; for $K > 2$ only the two-sided χ^2 is available.

4 Choice of the Weight Function

Weights tune power against specific alternatives; all have $W_k(t_j) = Y_{jk}W(t_j)$.

1. **Log-rank**: $W(t) = 1$; optimal against **proportional hazards** $h_1 = c_2 h_2 = \cdots = c_K h_K$.
2. **Gehan** (generalized Wilcoxon): $W(t) = Y(t)$, so $Z_k(\tau) = \sum_j (Y_j d_{jk} - d_j Y_{jk})$.
3. **Tarone–Ware**: $W(t) = f(Y(t))$, increasing f (original $f(u) = u^{1/2}$; also $\log u$); more weight to large risk sets.
4. **Peto–Peto**: $W(t) = \tilde{S}(t) = \prod_{t_j \le t}(1 - \frac{d_j}{Y_j+1})$ (pooled).
5. **Modified Peto–Peto** (Andersen, 1982): $W(t) = \tilde{S}(t)Y(t)/(Y(t)+1)$.
6. **Fleming–Harrington**: $W(t_j) = \hat{S}(t_{j-1})^p[1 - \hat{S}(t_{j-1})]^q$, $\hat{S}$ the pooled KM:
   - $p = q = 0$: log-rank;
   - $p > 0, q = 0$: sensitive to early differences;
   - $p = 0, q > 0$: sensitive to late differences.

Remark 9.8: Why the left-continuous $\hat{S}(t_{j-1})$

The weight at t_j must be *predictable* (known just before t_j), as martingale theory requires.

5 Catheter Placement in Kidney Dialysis Patients

Time to exit-site infection for catheters placed surgically (group 1) vs. percutaneously (group 2) (K&M Section 1.4). Surgery is better early, percutaneous later (about 0.8 vs. 0.2 near month 27): the KM curves *cross*, so the weight matters:

- early weighting (e.g. Gehan) may favour surgery;
- late weighting may favour percutaneous placement.

Relative weights (K&M Fig. 7.2): Gehan and Tarone–Ware decrease steeply, log-rank is flat, FH with $q > 0$ increase.

Example 9.9: Log-rank computation (K&M Table 7.2, selected rows)

t_j	Y_{j1}	d_{j1}	Y_{j2}	d_{j2}	Y_j	d_j	$Y_{j1}\frac{d_j}{Y_j}$	$d_{j1} - Y_{j1}\frac{d_j}{Y_j}$	$\frac{Y_{j1}}{Y_j}\left(1-\frac{Y_{j1}}{Y_j}\right)\frac{Y_j-d_j}{Y_j-1}d_j$
0.5	43	0	76	6	119	6	2.168	−2.168	1.326
1.5	43	1	60	0	103	1	0.417	0.583	0.243
2.5	42	0	56	2	98	2	0.857	−0.857	0.485
4.5	36	2	43	0	79	2	0.911	1.089	0.490
							⋮		
26.5	2	1	3	0	5	1	0.400	0.600	0.240
SUM		15		11		26	11.036	3.964	6.211

$Z_1(\tau) = 3.964$, $\hat{\sigma}_{11} = 6.211$, $\chi^2 = 2.53$, $p = 0.112$: not significant.

Example 9.10: Comparison of two-sample tests (K&M Table 7.3)

Test	$W(t_j)$	$Z_1(\tau)$	σ^2_{11}	χ^2	p-value
Log-rank	1	3.96	6.21	2.53	0.112
Gehan	Y_j	−9	38862	0.002	0.964
Tarone–Ware	$Y_j^{1/2}$	13.20	432.83	0.40	0.526
Peto–Peto	$\tilde{S}(t_j)$	2.47	4.36	1.40	0.237
Modified Peto–Peto	$\tilde{S}(t_j)Y_j/(Y_j+1)$	2.31	4.20	1.28	0.259
FH $p=0, q=1$	$1-\hat{S}(t_{j-1})$	1.41	0.21	9.67	0.002
FH $p=1, q=0$	$\hat{S}(t_{j-1})$	2.55	4.69	1.39	0.239
FH $p=1, q=1$	$\hat{S}(t_{j-1})[1-\hat{S}(t_{j-1})]$	1.02	0.11	9.83	0.002
FH $p=0.5, q=0.5$	$\hat{S}(t_{j-1})^{0.5}[1-\hat{S}(t_{j-1})]^{0.5}$	2.47	0.66	9.28	0.002
FH $p=0.5, q=2$	$\hat{S}(t_{j-1})^{0.5}[1-\hat{S}(t_{j-1})]^2$	0.32	0.01	8.18	0.004

Remark 9.11: Comments

- Gehan weights early comparisons heavily: *negative* $Z_1(\tau)$.
- FH tests with $q > 0$ are significant: the curves diverge for large t.
- Choose the weight *before* looking at the data.
- Log-rank is most popular, then Gehan.

6 Three-Group Comparison (Bone Marrow Transplant)

Disease-free survival after transplant: 38 ALL, 54 AML low-risk, 45 AML high-risk. Test $H_0 : S_1 = S_2 = S_3$ on $t \le \tau = 2204$ days.

Example 9.12: Log-rank test with $K = 3$

$Z_1(2204) = 2.148$, $Z_2(2204) = -14.966$,

$$\hat{\Sigma} = \begin{pmatrix} 15.9552 & -10.3451 \\ -10.3451 & 20.3398 \end{pmatrix}, \qquad \chi^2 = (2.148,\ -14.966)\,\hat{\Sigma}^{-1}\begin{pmatrix} 2.148 \\ -14.966 \end{pmatrix} = 13.8037.$$

On 2 df, $p \approx 0.001$.

Derivation: evaluating the quadratic form [Supplement]

$\det \hat{\Sigma} = 324.53 - 107.02 = 217.51$ and $\hat{\Sigma}^{-1} = \frac{1}{217.51}\begin{pmatrix} 20.3398 & 10.3451 \\ 10.3451 & 15.9552 \end{pmatrix}$. Then

$$\begin{aligned} \chi^2 &= \frac{1}{217.51}\Big[20.3398(2.148)^2 + 2(10.3451)(2.148)(-14.966) + 15.9552(-14.966)^2\Big] \\ &= \frac{93.85 - 665.14 + 3573.73}{217.51} \approx 13.80. \end{aligned}$$

Gehan $\chi^2 = 16.24$ ($p = 0.0003$); Tarone–Ware 15.65 (0.0004); FH $(1,0)$ 15.67 (0.0004); FH $(0,1)$ 6.11 (0.047); FH $(1,1)$ 9.93 (0.007): all indicate different curves.

7 Selected exercises

Motivation. The hypergeometric moments behind $\hat{\Sigma}$, the log-rank test as a score test, and invariance of χ^2 to the dropped group.

Exercise 9.1 (Hypergeometric moments)

$Y \geq 2$ at risk, Y_k in group k, and the d deaths a uniform random d-subset; d_k the deaths in group k. Prove the stated $\mathrm{E}(d_k)$, $\mathrm{Var}(d_k)$, $\mathrm{Cov}(d_k, d_{k'})$.

Solution [Supplement].

Step 1. With I_i indicating death of subject i: $P(I_i = 1) = d/Y$, $P(I_i = I_l = 1) = \binom{Y-2}{d-2}/\binom{Y}{d} = \dfrac{d(d-1)}{Y(Y-1)}$ $(i \neq l)$.

Step 2. So $\mathrm{Var}(I_i) = \frac{d}{Y}(1 - \frac{d}{Y})$ and

$$\mathrm{Cov}(I_i, I_l) = \frac{d(d-1)}{Y(Y-1)} - \frac{d^2}{Y^2} = \frac{d\{(d-1)Y - d(Y-1)\}}{Y^2(Y-1)} = -\frac{d(Y-d)}{Y^2(Y-1)} =: -c.$$

Step 3. $d_k = \sum_{i \in \text{group } k} I_i$: $\mathrm{E}(d_k) = dY_k/Y$ and

$$\mathrm{Var}(d_k) = Y_k \frac{d(Y-d)}{Y^2} - Y_k(Y_k - 1)c = \frac{Y_k d(Y-d)}{Y^2}\Big\{1 - \frac{Y_k - 1}{Y-1}\Big\} = d\,\frac{Y_k}{Y}\Big(1 - \frac{Y_k}{Y}\Big)\frac{Y-d}{Y-1}.$$

Step 4. For $k \neq k'$, over $Y_k Y_{k'}$ distinct pairs,

$$\mathrm{Cov}(d_k, d_{k'}) = -Y_k Y_{k'} c = -d\,\frac{Y_k Y_{k'}}{Y^2}\,\frac{Y-d}{Y-1}.$$

■

Exercise 9.2 (The one-sample log-rank test is a score test)

Let $T = \min(X, C)$, $\delta = I(X \leq C)$ with $X \perp\!\!\!\perp C$, X continuous with cumulative hazard H.
(a) Show that $\mathrm{E}(\delta) = \mathrm{E}\{H(T)\}$.
(b) In the exponential model $h \equiv \lambda$ (no truncation), show that the score statistic for $H_0 : \lambda = \lambda_0$, with the information estimated by $\sum_i T_i/\lambda_0$, equals the one-sample log-rank statistic $(O-E)^2/E$.

Solution [Supplement].

(a) With $G(s) = P(C \geq s)$, $H(T) = \int_0^\infty I(T \geq s)h(s)\,\mathrm{d}s$, Tonelli and independence,

$$\mathrm{E}\{H(T)\} = \int_0^\infty P(X \geq s)G(s)h(s)\,\mathrm{d}s = \int_0^\infty f(s)G(s)\,\mathrm{d}s = P(X \leq C) = \mathrm{E}(\delta),$$

as $P(X \geq s)h(s) = f(s)$. (The counting process minus its compensator has mean zero.)

(b)

Step 1. With $O = \sum_i \delta_i$, $\Sigma T = \sum_i T_i$: $\ell = O\log\lambda - \lambda\,\Sigma T$, $U(\lambda) = O/\lambda - \Sigma T$, $\mathcal{I}(\lambda) = \mathrm{E}(O)/\lambda^2$.

Step 2. By (a), $\mathrm{E}(O) = \lambda_0 \mathrm{E}(\Sigma T)$ under H_0, so $\mathcal{I}(\lambda_0)$ is estimated by $\Sigma T/\lambda_0$.

Step 3. $E = \sum_i H_0(T_i) = \lambda_0 \Sigma T$ ($L_i = 0$), so

$$\frac{U(\lambda_0)^2}{\Sigma T/\lambda_0} = \frac{(O - \lambda_0\Sigma T)^2/\lambda_0^2}{\Sigma T/\lambda_0} = \frac{(O-E)^2}{\lambda_0 \Sigma T} = \frac{(O-E)^2}{E}.$$

■

Exercise 9.3 (The χ^2 does not depend on the dropped group)

Let $\mathbf{Z} = (Z_1, \ldots, Z_K)'$ and $\hat{\Sigma}$ be as in this lecture. For each k let $\mathbf{Z}_{(-k)}$ and $\hat{\Sigma}_{(-k)}$ be obtained by deleting the k-th entry, respectively the k-th row and column, and assume $\hat{\Sigma}_{(-K)}$ is invertible.
(a) Show that $\hat{\Sigma}\mathbf{1} = \mathbf{0}$.
(b) Show that $\hat{\Sigma}_{(-k)}$ is invertible and $\mathbf{Z}'_{(-k)}\hat{\Sigma}^{-1}_{(-k)}\mathbf{Z}_{(-k)} = \mathbf{Z}'_{(-K)}\hat{\Sigma}^{-1}_{(-K)}\mathbf{Z}_{(-K)}$ for every k.

Solution [Supplement].

(a) At a death time with $\pi_k = Y_{jk}/Y_j$, $w = W(t_j)^2 d_j(Y_j - d_j)/(Y_j - 1)$, the contribution to $\sum_{k'} \hat{\sigma}_{kk'}$ is

$$w\Big\{\pi_k(1-\pi_k) - \sum_{k' \neq k} \pi_k \pi_{k'}\Big\} = w\,\pi_k\Big\{1 - \sum_{k'} \pi_{k'}\Big\} = 0.$$

(b)

Step 1. It suffices to compare $k = 1$ with K. With P_k deleting coordinate k, $\mathbf{Z}_{(-k)} = P_k\mathbf{Z}$, $\hat{\Sigma}_{(-k)} = P_k \hat{\Sigma} P_k'$.

Step 2. On $V = \{\mathbf{z} : \mathbf{1}'\mathbf{z} = 0\}$, $P_1\mathbf{z} = M P_K \mathbf{z}$ with $M : (z_1, \ldots, z_{K-1}) \mapsto (z_2, \ldots, z_{K-1}, -\sum_{k<K} z_k)$, invertible.

Step 3. $\mathbf{Z} \in V$, and by (a) and symmetry every column of $\hat{\Sigma}$ lies in V. Applying Step 2 to the columns of $\hat{\Sigma}P_1'$, then of $\hat{\Sigma}P_K'$,

$$P_1\hat{\Sigma}P_1' = MP_K\hat{\Sigma}P_1' = M\big(P_1\hat{\Sigma}P_K'\big)' = M\big(MP_K\hat{\Sigma}P_K'\big)' = M\hat{\Sigma}_{(-K)}M'.$$

Step 4. So $\hat{\Sigma}_{(-1)} = M\hat{\Sigma}_{(-K)}M'$ is invertible, and with $\mathbf{Z}_{(-1)} = M\mathbf{Z}_{(-K)}$,

$$\mathbf{Z}'_{(-1)}\hat{\Sigma}^{-1}_{(-1)}\mathbf{Z}_{(-1)} = \mathbf{Z}'_{(-K)}M'(M')^{-1}\hat{\Sigma}^{-1}_{(-K)}M^{-1}M\mathbf{Z}_{(-K)} = \mathbf{Z}'_{(-K)}\hat{\Sigma}^{-1}_{(-K)}\mathbf{Z}_{(-K)}.$$

■